

The Oxford Implementation Index: a new tool for incorporating implementation data into systematic reviews and meta-analyses

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Abstract

Objectives: This article presents a new tool that helps systematic reviewers to extract and compare implementation data across primary trials. Currently, systematic review guidance does not provide guidelines for the identification and extraction of data related to the implementation of the underlying interventions.

Study Design and Setting: A team of systematic reviewers used a multistaged consensus development approach to develop this tool. First, a systematic literature search on the implementation and synthesis of clinical trial evidence was performed. The team then met in a series of subcommittees to develop an initial draft index. Drafts were presented at several research conferences and circulated to methodological experts in various health-related disciplines for feedback. The team systematically recorded, discussed, and incorporated all feedback into further revisions. A penultimate draft was discussed at the 2010 Cochrane–Campbell Collaboration Colloquium to finalize its content.

Results: The Oxford Implementation Index provides a checklist of implementation data to extract from primary trials. Checklist items are organized into four domains: *intervention design*, *actual delivery* by trial practitioners, *uptake* of the intervention by participants, and *contextual factors*. Systematic reviewers piloting the index at the Cochrane–Campbell Colloquium reported that the index was helpful for the identification of implementation data.

Conclusion: The Oxford Implementation Index provides a framework to help reviewers assess implementation data across trials. Reviewers can use this tool to identify implementation data, extract relevant information, and compare features of implementation across primary trials in a systematic review. The index is a work-in-progress, and future efforts will focus on refining the index, improving usability, and integrating the index with other guidance on systematic reviewing. © 2013 Elsevier Inc. All rights reserved.

Keywords: Systematic review; Implementation; Heterogeneity; Generalizability; Randomized controlled trial; Reporting guideline

Conflict of interest: The Oxford Implementation Index was developed by a team of systematic reviewers at the Centre for Evidence-Based Intervention at Oxford University. The team members are affiliated with the Cochrane and Campbell Collaborations and independently conduct randomized controlled trials in areas of public health, psychiatry, and social welfare. P.M. is the author of over a dozen Cochrane Reviews and was, until recently, cochair of the Campbell Collaboration Social Welfare Group. He led this team and managed the project. P.M., E.M.-W., D.O., F.G., and K.U. contributed to the conception of the index and initial project planning. A systematic literature search was developed and conducted by K.U. and P.M. The index was developed by all the authors, led by P.M. and K.U. Piloting was led by P.M. and facilitated by F.G., E.M.-W., D.O., and K.U. K.U., E.M.-W., and P.M. initially evaluated the index using their own reviews. The Delphi panel was run by P.M. and K.U.; all the authors discussed feedback and, led by P.M., refined the index. All the authors contributed to the writing and revision of the article, led by K.U., E.M.-W., and P.M. The index was the focus of a doctoral dissertation by K.U.

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